Lecture 17

Linear Higher Order Equations

We have so far explored linear 1st and 2nd-order equations in detail. We can approach linear higher order equations in a similar fashion. By first considering homogeneous constant coefficient equations and then building upon this to find particular solutions using similar techniques to undetermined coefficients and variation of parameters.

Consider the linear n-th order equation

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)} + \cdots a_1(t)y'(t) + a_0(t)y(t) = f(t), \quad a_n(t) \neq 0.$$

We can put it in **standard form** as divide by $a_n(t)$

$$1 \cdot y^{(n)} + p_1(t)y^{(n-1)} + \cdots p_{n-1}(t)y' + p_n(t)y = g(t) \leftarrow \frac{f(t)}{a_n(t)}$$

and we have a similar situation about the existence of solutions as for 2nd-order equations.

Theorem (Existence and Uniqueness of Solutions to Linear n-th Order Equations)**:** Suppose that $p_1(t), \ldots, p_n(t)$, and $g(t)$ are continuous on an interval I containing t_0. Then for any choice of initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_1, \ldots, \quad y^{(n-1)}(t_0) = y_{n-1}$$

there exists a unique solution $y(t)$ on the interval I to the IVP

$$\begin{cases} y^{(n)}(t) + p_1(t)y^{(n-1)}(t) + \cdots + p_n(t)y(t) = g(t) \\ y(t_0) = y_0, \quad y'(t_0) = y_1, \ldots, \quad y^{(n-1)}(t_0) = y_{n-1} \end{cases}$$

Example: On what interval is the equation

$$ty''' - 2\sin(t)y'' + \sqrt{t+4}y = \ln(t)$$

guaranteed a unique solution by the E-U Theorem at $t = +1$? $t=1$

standard form: divide by t

$$y''' - 2\frac{\sin(t)}{t}y'' + \frac{\sqrt{t+4}}{t}y = \frac{\ln(t)}{t}$$

$p_1(t) = -\frac{2\sin(t)}{t}$, $p_2(t) = 0$, $p_3(t) = \frac{\sqrt{t+4}}{t}$, $g(t) = \frac{\ln(t)}{t}$

cont: $(-\infty,0)\cup(0,\infty)$, $(-\infty,\infty)$, $[-4,0)\cup(0,\infty)$, $(0,\infty)$

simultaneously cont. on: $(0,\infty)$.

For $t=1$, we guaranteed a unique solution on $(0,\infty)$.

Just as with linear 1st and 2nd-order equations, linear nth-order homogeneous equations have exactly n linearly independent solutions and a general homogeneous solution is of the form

$$\boxed{y_h = c_1y_1 + c_2y_2 + \cdots c_ny_n}$$

for some collection $y_1, y_2, \ldots, y_n$ of linearly independent solutions.

$\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_1'y_2$

How can we tell that a collection of solutions is linearly independent. In the 2nd-order case we used the $\boxed{\text{Wronskian } W[y_1,y_2] = y_1y_2' - y_1'y_2}$ as the determinant of a 2 x 2 matrix. Before we can make the analogous definition for the n-th order case we must first discuss higher order determinants.

Consider a 3 x 3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{minor}(A)_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

the **ijth-minor**, minor$(A)_{ij}$, is the determinant of the 2 x 2 matrix obtained by deleting the ith row and jth column from the matrix A.

Example: Consider the matrix

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Find minor$(B)_{11}$ and minor$(B)_{23}$.

$$\text{minor}(B)_{11} = \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = 5(9) - 8(6) = 45 - 48 = -3$$

$$\text{minor}(B)_{23} = \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 1(8) - 7(2) = 8 - 14 = -6$$

row column

$$C_{11} = (-1)^{1+1}\text{minor}(B)_{11} = (-1)^2(-3) = -3$$

$$C_{23} = (-1)^{2+3}\text{minor}(B)_{23} = (-1)^5(-6) = 6$$

Next, for the matrix A the **ijth-cofactor**, C_{ij}, is defined as

$$C_{ij} = (-1)^{i+j}\text{minor}(A)_{ij}.$$

With this we can define the determinant of a 3 x 3 matrix:

Definition: Given the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

we define the **determinant** of A as

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

cofactor expansion

Note: In this definition we chose to "expand" the cofactor sum along the first row, but you can define the determinant in an equivalent way by expanding along any row or column with the appropriate cofactors.

The determinant of any n x n matrix can now be stated: recursively

Definition: Given the n x n matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots a_{nn} & \end{bmatrix}$$

the **determinant** of A is defined as

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots a_{1n}C_{1n} = \sum_{i=1}^{n} a_{1i}C_{1i}$$

where C_{ij} is the ij^{th}-cofactor of the matrix A.

Note: We will use vertical bars to denote the determinant (just as we did for the 2 x 2 case of the Wronskian and variation of parameters).

Example: Find the determinant of

$C_{ij} = (-1)^{i+j} \text{minor}(A)_{ij}$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\det(B) = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1 \cdot C_{11} + 2 \cdot C_{12} + 3 \cdot C_{13}$$

$$= 1 \cdot (-1)^{1+1} \text{minor}(B)_{11} + 2 \cdot (-1)^{1+2} \text{minor}(B)_{12} + 3(-1)^{1+3} \text{minor}(B)_{13}$$

$$= 1 \cdot (-1)^2 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + 2(-1)^3 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3(-1)^4 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

$$= (-3) + 2(-1)(4(9) - 7(6)) + 3(1)(4(8) - 7(5))$$

$$= -3 - 2(36 - 42) + 3(32 - 35) = -3 - 2(-6) + 3(-3)$$

$$= -3 + 12 - 9 = \boxed{0}$$

Now we can define the Wronskian of a larger collection of functions or solutions using determinants:

Definition: Let $y_1, \ldots, y_n$ be any n functions that are $(n-1)$ times differentiable. The function

$$W[y_1, y_2, \ldots, y_n](t) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

is called the **Wronskian** of $y_1, \ldots, y_n$.

$y_1, y_2, y_3 \quad W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$

Example: Find the Wronskian of the collection of functions $y_1 = t$, $y_2 = e^t$, and $y_3 = \ln(t)$

$$W[y_1, y_2, y_3](t) = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} t & e^t & \ln(t) \\ 1 & e^t & \frac{1}{t} \\ 0 & e^t & -\frac{1}{t^2} \end{vmatrix}$$

$$= t \cdot C_{11} + e^t \cdot C_{12} + \ln(t) C_{13}$$

$$= t \cdot (-1)^{1+1} \begin{vmatrix} e^t & \frac{1}{t} \\ e^t & -\frac{1}{t^2} \end{vmatrix} + e^t (-1)^{1+2} \begin{vmatrix} 1 & \frac{1}{t} \\ 0 & -\frac{1}{t^2} \end{vmatrix} + \ln(t)(-1)^{1+3} \begin{vmatrix} 1 & e^t \\ 0 & e^t \end{vmatrix}$$

$$= t(1)\left(-e^t \cdot \frac{1}{t^2} - e^t \cdot \frac{1}{t}\right) + e^t \underline{(-1)\left(-\frac{1}{t^2} - 0\right)} + \ln(t)(1)(e^t - 0)$$

$$= -\frac{e^t}{t} - e^t + \frac{e^t}{t^2} + e^t \ln(t) \quad \text{to } (0, \infty) \text{ except } t = 1.$$

There is a similar result about linear dependence and independence for homogeneous solutions to linear nth-order constant coefficient equations:

Theorem: Suppose that $y_1, y_2, \ldots y_n$ are solutions to the linear nth-order homogeneous equation

$$y^{(n)} + p_1(t)y^{(n-1)} + \cdots + p_{n-1}(t)y' + p_n(t)y = 0,$$

where $p_1, p_2, \ldots, p_n$ are continuous on an interval I. If $W[y_1, y_2, \ldots y_n](t) = 0$ for some value of t in the interval I, then the functions $y_1, y_2, \ldots, y_n$ are linearly dependent on the interval I. Otherwise, they are linearly independent.

To find solutions to linear higher order equations, we will first look at homogeneous constant coefficient equations before proceeding to non-homogeneous equations and then variable coefficient equations.

It will be useful to come up with an alternate notation to write our equations with.

Definition: Suppose that $y(t)$ is a differentiable function. The **differential operator** D is defined as the following:

$$D(y) = \frac{dy}{dt}, \quad D^n(y) = \frac{d^n y}{dt^n}$$

Given a homogeneous constant coefficient equation

$$a_n y^{(n)} + \cdots a_1 y' + a_0 y = 0$$

$D^n(y)$

$D(y)$

we can write it equivalently using the differential operator D as

$$[a_n D^n + \cdots a_1 D + a_0](y) = 0$$

Example: Write the following equations using the differential operator D

- $2y'' + 3y' + y = 0 \Rightarrow 2D^2(y) + 3D(y) + y = 0$

  $\Rightarrow [2D^2 + 3D + 1](y) = 0$

- $3y''' + 2y' = 0 \Rightarrow 3D^3(y) + 2D(y) = 0$

  $\Rightarrow [3D^3 + 2D](y) = 0$

- $y^{(4)} - 16y = 0 \Rightarrow D^4(y) - 16y = 0$

  $\Rightarrow [D^4 - 16](y) = 0$

Note: Rewriting a 2nd-order homogeneous constant coefficient equation

$$ay'' + by' + cy = 0 \Rightarrow aD^2(y) + bD(y) + cy = 0$$

using the differential operator D gives

$$[aD^2 + bD + c]y = 0$$

which extracts the characteristic equation by replacing D with r.

char eqn: $ar^2 + br + c$

In general, for a homogeneous constant-coefficient equation

$$a_n y^{(n)} + \cdots a_1 y' + a_0 y = 0 \quad \Rightarrow \quad [a_n D^n + \cdots a_1 D + a_0](y) = 0$$

the **characteristic equation** (or **auxiliary equation**) is defined as

$$a_n r^n + \cdots a_1 r + a_0 = 0$$

Note: We can arrive at the same characteristic equation by making the substitution $y = e^{rx}$ as we did for the 2nd-order case.

Example: Write the characteristic equation for

$D^2(D+1) + (D+1)$
$[(D^2+1)(D+1)](y) = 0$

- $y''' + y'' + y' + y = 0$ $\Rightarrow [D^3 + D^2 + D + 1](y) = 0$

  char eqn: $r^3 + r^2 + r + 1 = 0$
  $\Rightarrow r^2(r+1) + r + 1 = (r^2+1)(r+1) = 0$

- $y^{(5)} + y^{(3)} + y' = 0$ $\Rightarrow [D^5 + D^3 + D](y) = 0$

  char eqn: $r^5 + r^3 + r = 0$

The characteristic equation consists of a polynomial equal to zero and by the Fundamental Theorem of Algebra, we are guaranteed n real or complex roots to the equation. Just as in the 2nd-order case, we have three possibilities for solution types based upon the characteristic equation:

Case 1: n distinct real roots $r_1 \neq r_2 \neq \cdots \neq r_n$.

In this case, the general solution is $y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t} + \cdots c_n e^{r_n t}$.

One thing to note is that if the characteristic equation can be written as

$$(r - r_1)(r - r_2) \cdots (r - r_n) = 0$$

Then the original equation can be written as

$$[(D - r_1)(D - r_2) \cdots (D - r_n)](y) = 0$$

and in fact, the factoring of the characteristic equation is equivalent to factoring the operator form of the equation.

multiplicity

Case 2: repeated real roots

ex: $(r-2)^2(r-3)^2 = 0$

2, 2 3, 3

If there is a repeated real root r_1, it is repeated a certain number of times, say m. We call m the **multiplicity** of r_1. For a root r_1 of multiplicity m, the portion of the solution that comes from the root r_1 is

$$\underbrace{c_1 e^{r_1 t} + c_2 t e^{r_1 t} + \cdots c_m t^{m-1} e^{r_1 t}}_{t^0 \quad t^1 \quad \cdots \quad t^{m-1}}$$

If there is a factor $(r - r_1)^m$ in the characteristic equation, the we must have a factor of $(D - r_1)^m$ in the operator form of the equation.

Case 3: complex conjugate pairs $r = \alpha + i\beta$

In this case, the portion of the solution from the root $\alpha + i\beta$ is

$$c_1 e^{\alpha t}\cos(\beta t) + c_2 e^{\alpha t}\sin(\beta t)$$

If there is a root $\alpha + i\beta$ then $\alpha - i\beta$ is also a root and so there is a factor of

$$(r - (\alpha + i\beta))(r - (\alpha - i\beta)) = r^2 - 2\alpha r + (\alpha^2 + \beta^2) = (r - \alpha)^2 + \beta^2$$

$\alpha + i\beta$

Then we must have a factor of $((D - \alpha)^2 + \beta^2)$ in the operator form of the equation.

Note: There may be repeated complex roots in the characteristic equation in which you must then use Case 2 and Case 3 in conjunction to write out a solution.

Example: Write the general solution to the following equations:

1. $y^{(4)} - y^{(3)} - 3y'' + 5y' - 2y = 0$

$D^4(y) - D^3(y) - 3D^2(y) + 5D(y) - 2y = 0$

$\Rightarrow [D^4 - D^3 - 3D^2 + 5D - 2](y) = 0$ $\Rightarrow$ char eqn: $r^4 - r^3 - 3r^2 + 5r - 2 = 0$

rational roots theorem
integer coefficients
$\Rightarrow$ rational root is of the form $\pm \dfrac{\text{factors of constant}}{\text{factors of leading}}$

possible roots: $\pm 1, \pm 2$

$r = 1$: $1^4 - 1^3 - 3(1)^2 + 5(1) - 2 = 1 - 1 - 3 + 5 - 2 = 0$ ✓

$$\begin{array}{r} r^3 \quad - 3r + 2 \\ r - 1 \overline{\big) r^4 - r^3 - 3r^2 + 5r - 2} \\ -(r^4 - r^3) \\ -2(-3r^2 + 3r) \\ -(2r - 2) \\ 2r - 2 \\ 0 \end{array}$$

$(r-1)(r^3 - 3r + 2)$ ← factor of $r - 1$

$$\begin{array}{r} r^2 + r - 2 \\ r - 1 \overline{\big) r^3 - 3r + 2} \\ -(r^3 - r^2) \\ r^2 - 3r \\ -(r^2 - r) \\ -2r + 2 \\ -2r + 2 \\ 0 \end{array}$$

$= (r-1)(r-1)(r^2 + r - 2)$

$= (r-1)(r-1)(r-1)(r+2)$

$\Rightarrow (r-1)^3(r+2) = 0$ $r = 1, 1, 1, -2$

$[(D-1)^3(D+2)](y) = 0$

$y_h = (c_1 e^t + c_2 t e^t + c_3 t^2 e^t) + c_4 e^{-2t}$

$$\frac{D^2+2D+1}{(D+1)^2} - 1$$

2. $[(D-1)^2(D+3)(D^2+2D+5)^2](y) = 0$ → char eqn: $(r-1)^2(r+3)(r^2+2r+5)^2 = 0$

$\Rightarrow [(D-1)^2(D+3)((D+1)^2+4)^2](y) = 0$

$(D-\bar{\alpha})^2 + \beta^2$

$r=1$, $r=-3$, $r=-1+2i, -1+2i$

repeated

$$y_h = (c_1 e^t + c_2 t e^t) + c_3 e^{-3t} + [c_4 e^{-t}\cos(2t) + c_5 e^{-t}\sin(2t)] + t[c_6 e^{-t}\cos(2t) + c_7 e^{-t}\sin(2t)]$$

3. $y''' - y'' + 2y = 0$ → $[D^3 - D^2 + 2](y) = 0$

factor

rational roots theorem: $\frac{\pm \text{factors of } 2}{\text{factors of } 1} = \pm 1, \pm 2$

$D=1$: $1^3 - 1^2 + 2 = 2 \neq 0$

$D=-1$: $(-1)^3 - (-1)^2 + 2 = -1 - 1 + 2 = 0$ ✓ root of -1

$$\begin{array}{r} D^2 - 2D + 2 \\ D+1 \overline{\big) D^3 - D^2 + 2} \\ -(D^3 + D^2) \\ -2D^2 \\ -(-2D^2 - 2D) \\ +2D + 2 \\ -(+2D + 2) \\ 0 \end{array}$$

$\Rightarrow [(D+1)(D^2 - 2D + 2)](y) = 0$

complete the square

$[(D+1)((D-1)^2 + 1)](y) = 0$

-1 $\quad 1+i, 1-i$

$$\boxed{y_h = c_1 e^{-t} + c_2 e^t\cos(t) + c_3 e^t\sin(t)}$$

$$\frac{D^2 - 2D + 1}{(D-1)^2} - 1$$